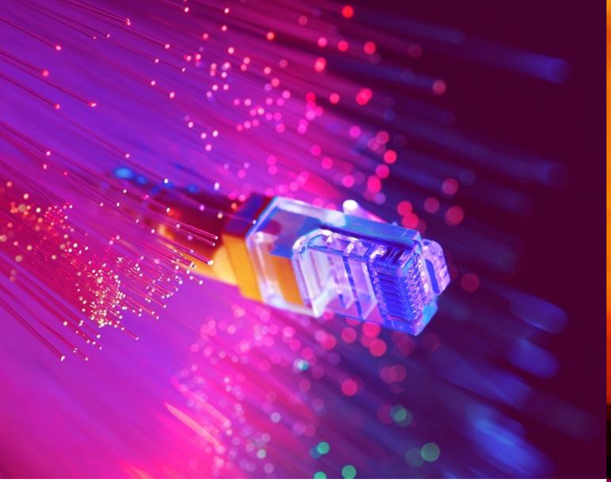
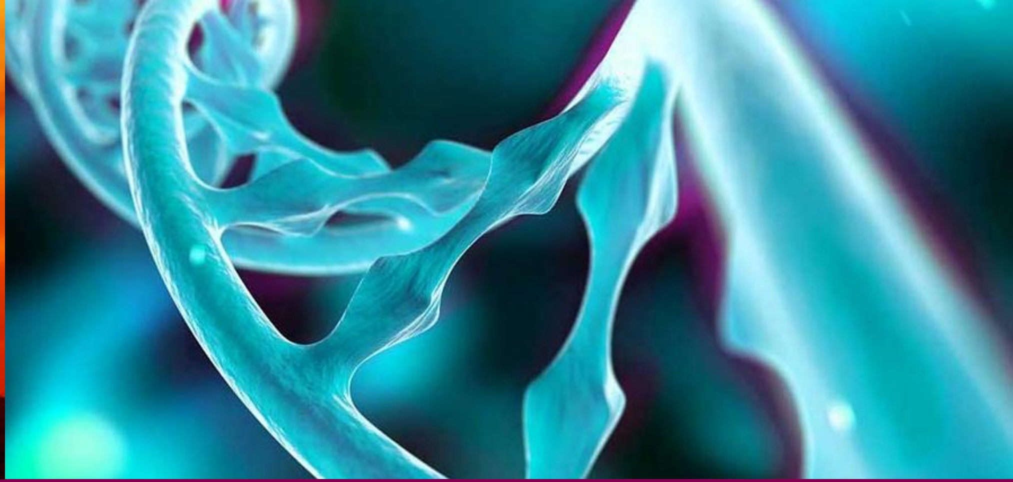




Bioinformatics data analysis using Chipster


CSC – Suomalainen tutkimuksen, koulutuksen, kulttuurin ja julkishallinnon ICT-osaamiskeskus

You will learn

- What Chipster is
- How the user interface works
- How to import data
- How to store and share your analysis
- How to speed up your analysis
    - run the same analysis step for all the samples simultaneously
- How to get help
- Where to find training materials

Chipster: User-friendly analysis software for high-throughput data

- Over 500 analysis tools, inc tools for single-cell RNA-seq and Visium data
  - Command line tools, R/Bioconductor packages
  - Free, open source software
- Users can share analysis sessions, analysis metadata is tracked
- Training resources
  - Course material (slides, videos, exercises, data sets) available
  - Training accounts available, email chipster@csc.fi
- Technical
  - Angular based web app and cloud native scalable backend
  - Ansible playbooks and Helm templates for setting up a K3s container orchestration system and Chipster containers on a virtual machine or on a physical server
- https://chipster.csc.fi/

https://chipster.csc.fi/

Chipster

Open source platform for data analysis

- Home
- Getting access
- Manual
- Tutorial videos
- Course material
- Cite
- Contact

Welcome to Chipster

Chipster is a user-friendly analysis software for high-throughput data such as Visium, single-cell and bulk RNA-seq. Chipster provides a web interface to over 500 analysis tools, and the actual analysis jobs run on the server side making use of CSC's computing environment (watch introduction video).

If you would like to use Chipster hosted by CSC, you need a user account. Please note that Chipster is also available for local server installation free of charge.

Launch Chipster

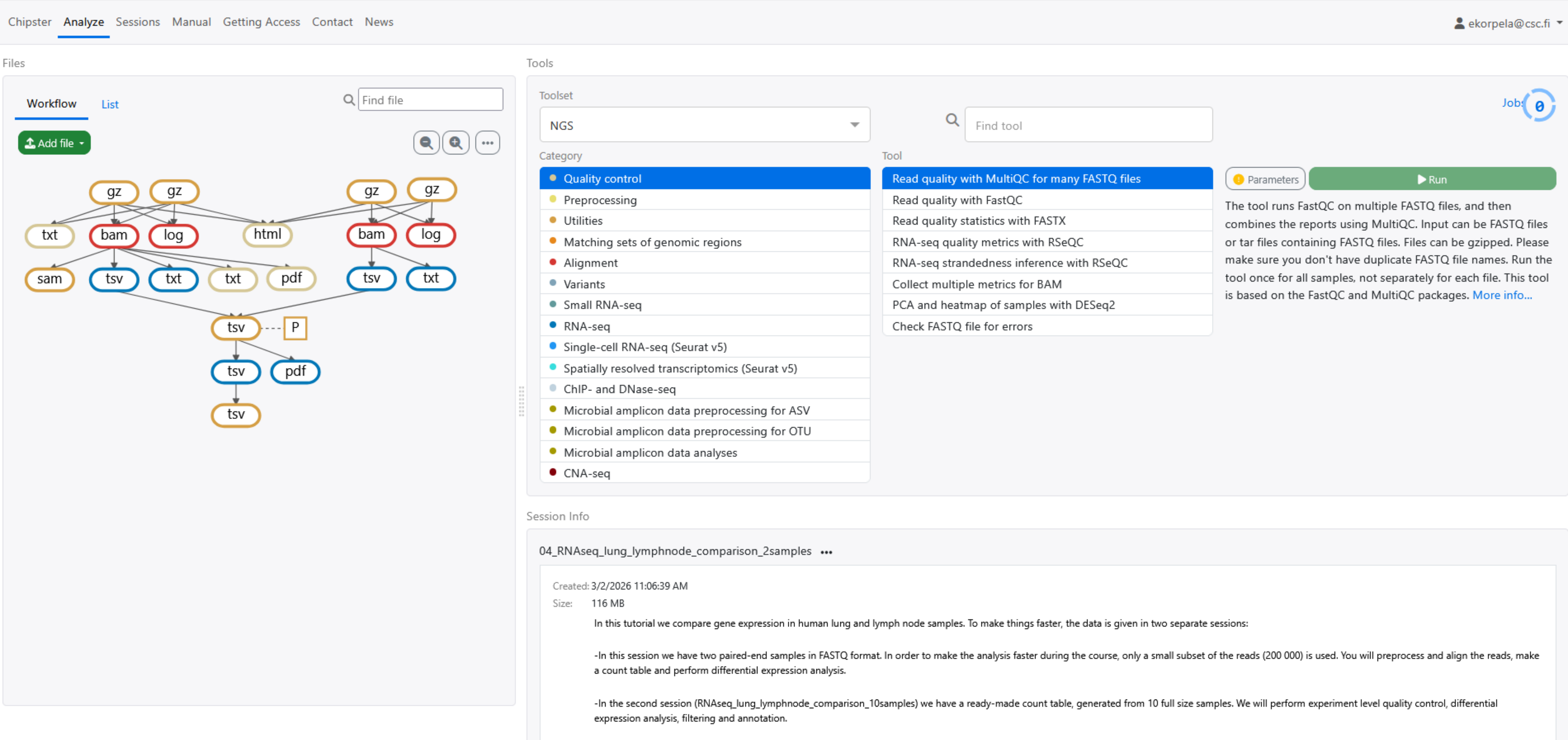
Training:

- 3.-4.3.2026 Single-cell RNA-seq data analysis
- 14.-15.4.2026 ASV-based microbial community / environmental DNA analysis
- MOOC Single-cell RNA-seq data analysis using Chipster
- MOOC Spatially resolved transcriptomics data analysis using Chipster

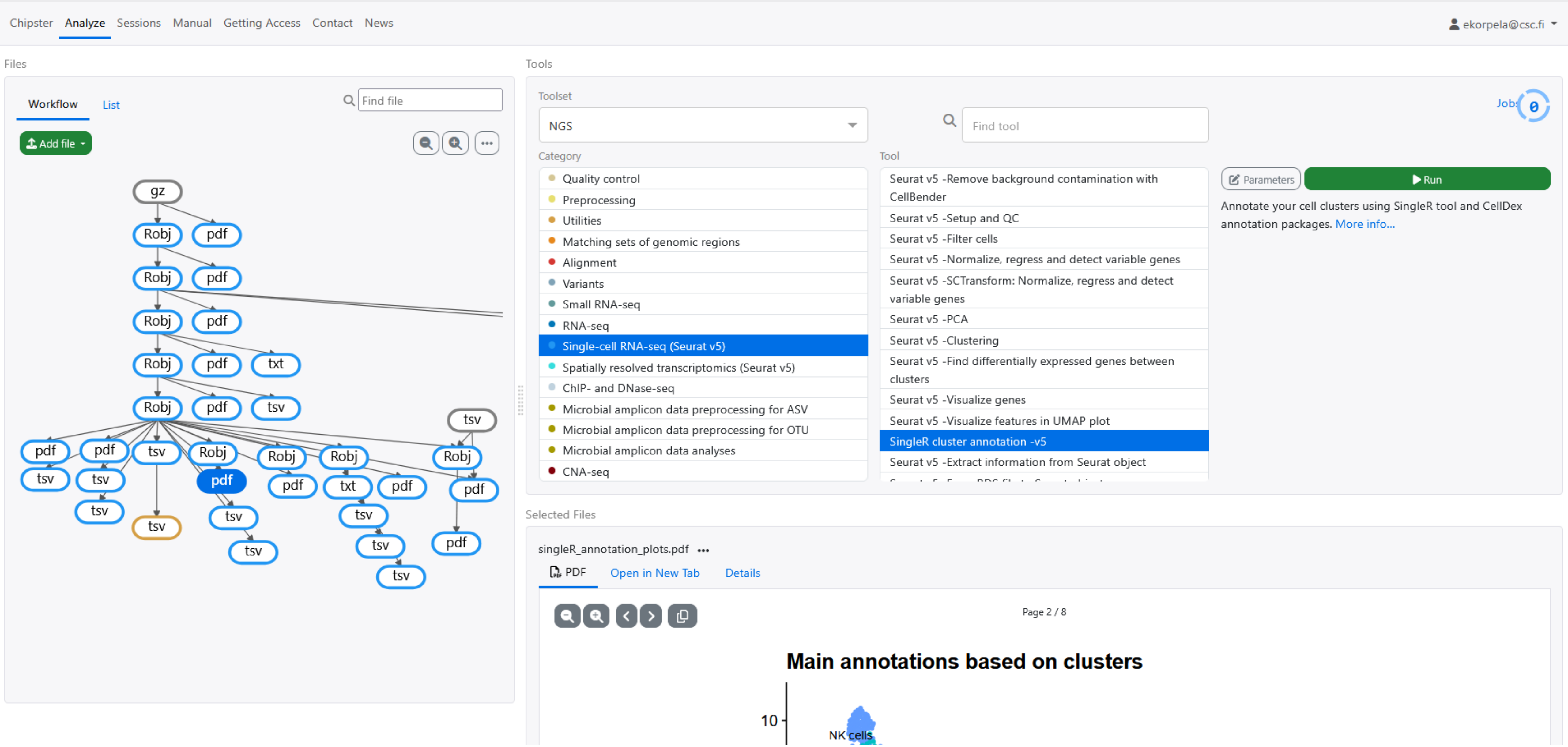
News and resources:

- video tour of Chipster Web app
- ASV-based microbial community analysis using DADA2: Tutorial videos
- Instructions for moving data from Puhti to Chipster
- Video on how to convert tables to Chipster format and create phenodata file

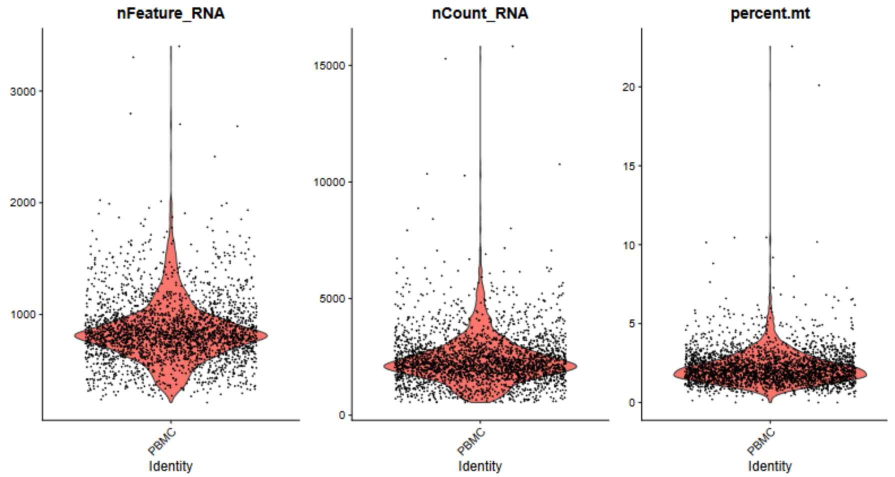
Chipster user interface (chipster.2.rahtiapp.fi)

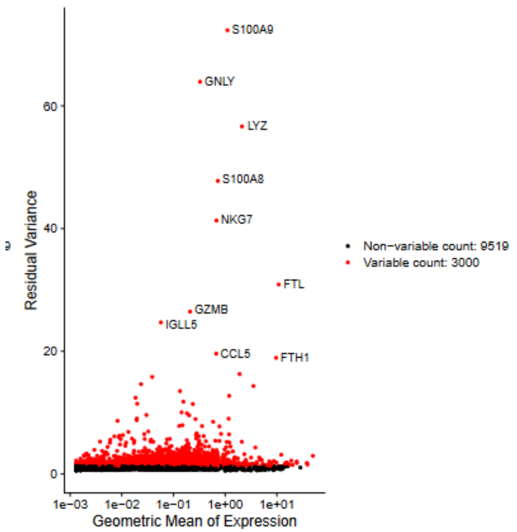


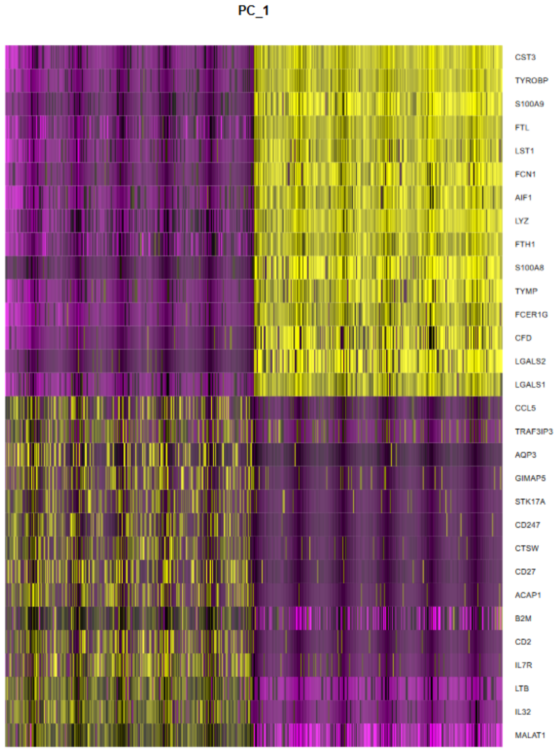
Chipster GUI, continued

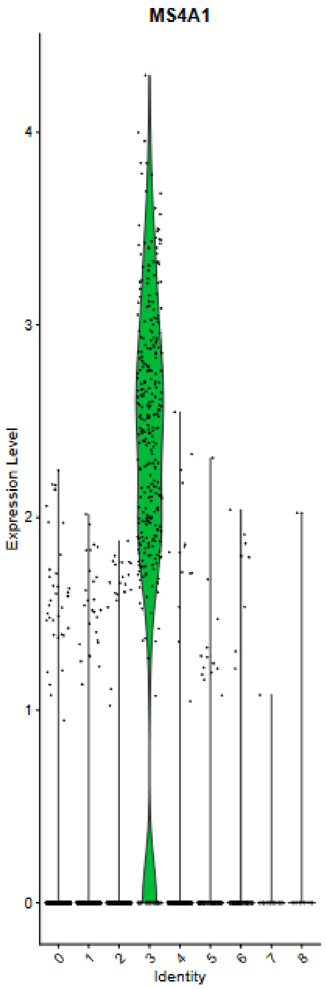


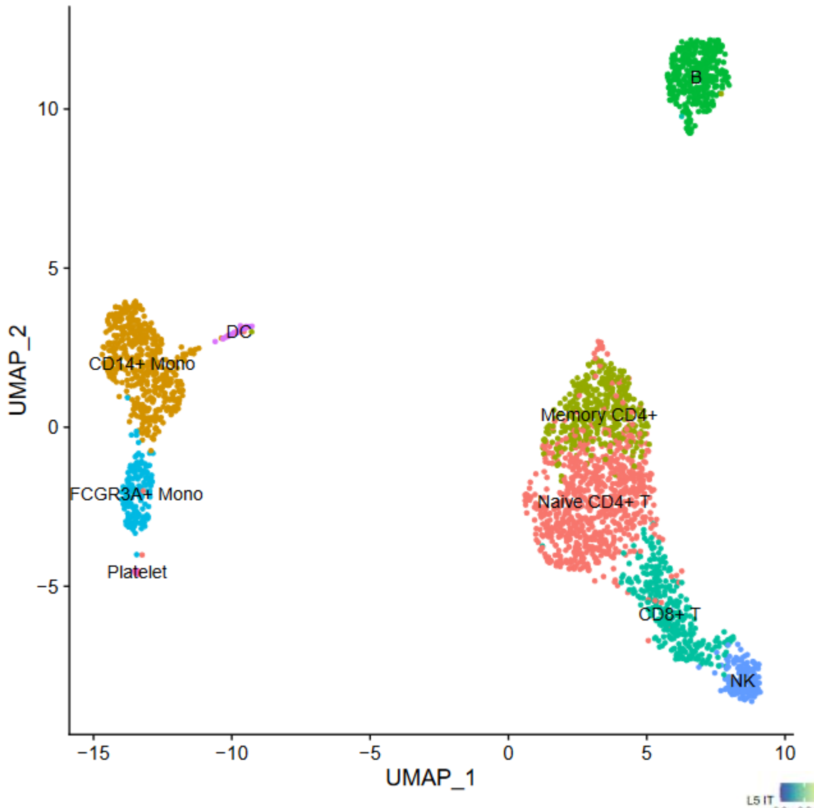
Chipster visualizations

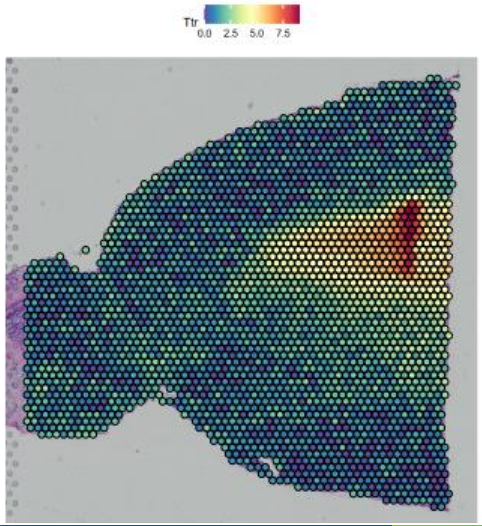


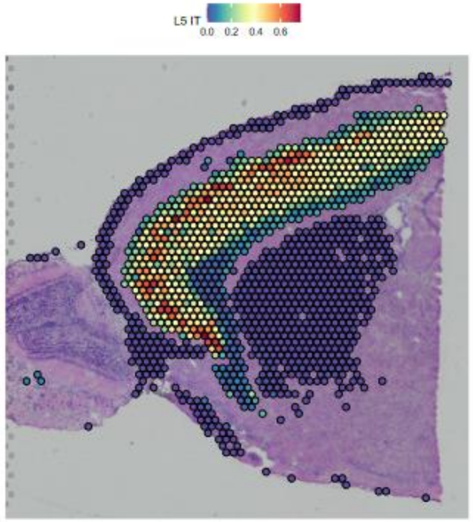












Options for importing data to Chipster

- ”Add file” button
  - Upload files
  - Upload folder
  - Download from URL

- Sessions tab
  - Import session file

- Tools
  - Import from Illumina BaseSpace
    - Utilities / Retrieve data from Illumina BaseSpace
    - Access token needed
  - Import from SRA database
    - Utilities / Retrieve FASTQ or BAM files from SRA
  - Import from Ensembl database
    - Utilities / Retrieve data for a given organism in Ensembl
  - Import from URL
    - Utilities / Download file from URL directly to server

Analysis sessions

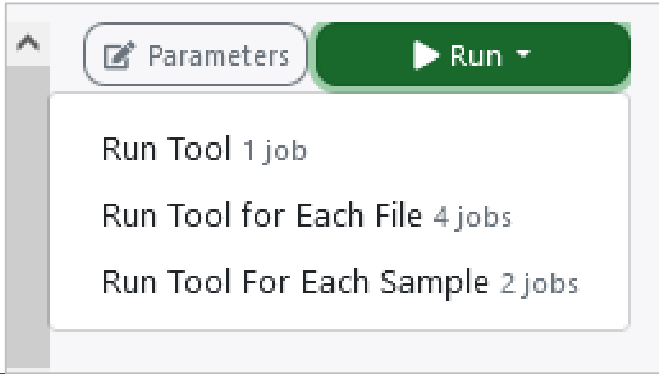
- Your analysis is saved automatically in the cloud
  - Session includes all the files, their relationships and metadata (what tool and parameters were used to produce each file).
  - Session is a single .zip file.
  - Note that cloud sessions are not stored forever! Remember to download the session when ready.
- You can share sessions with other Chipster users
  - You can give either read-only or read-write access
- If your analysis job takes a long time, you don’t need to keep Chipster open:
  - Wait that the data transfer to the server has completed
  - Close Chipster
  - Open Chipster later and the results will be there

Running many analysis jobs at the same time

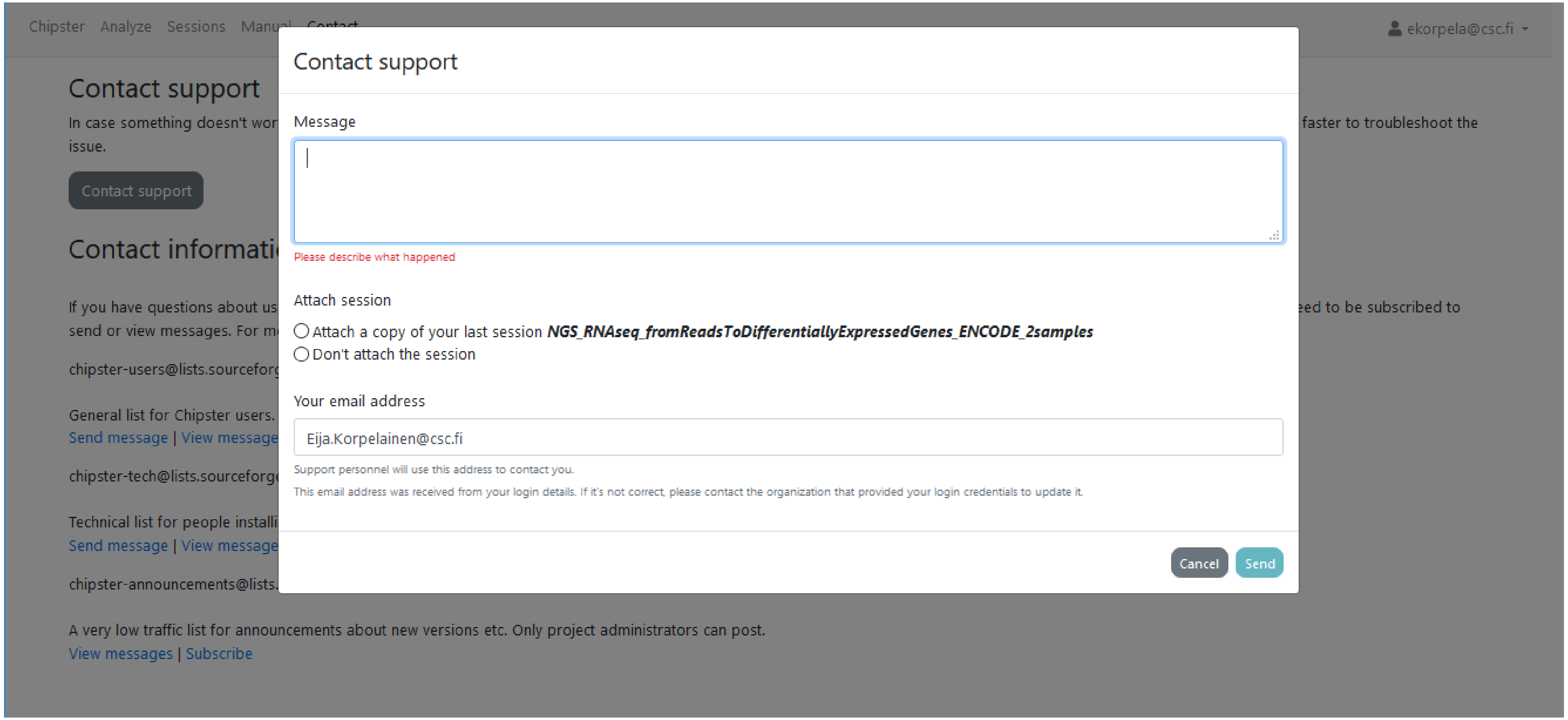
- You can have many analysis jobs running at the same time
  - No need to wait that one finishes before starting a new one

Run button gives several options:

- Run tool
  - Runs the selected analysis tool once
- Run tool for each file
  - Runs the selected analysis tool for each of the input files individually
- Run tool for each <u>sample</u>
  - If you have grouped paired end FASTQ files to samples using the Define samples –option, you can run the selected analysis tool for the input files in a sample specific manner.



Problems? Send us a support request
-request includes the error message and link to analysis session (optional)



More info

- chipster@csc.fi
- http://chipster.csc.fi
- https://chipster.2.rahtiapp.fi/manual/courses.html
- Chipster tutorials in YouTube